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AI-Driven Phishing Website Detection Using Machine Learning

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ABSTRACT

Phishing attacks remain one of the most persistent and rapidly evolving threats in cybersecurity. As attackers continuously develop new techniques to deceive users, traditional detection methods such as blacklists and rule-based filters are no longer sufficient. This research paper proposes an AI-driven phishing website detection system powered by advanced machine learning algorithms. The study utilizes lexical, host-based, and securityrelated URL features to classify websites as phishing or legitimate. Experimental analysis demonstrates that models such as LightGBM, XGBoost, and Random Forest significantly improve identification accuracy. The findings indicate that AI-based URL analysis can strengthen modern cybersecurity infrastructures.

KEYWORDS : Phishing Detection, Machine Learning, Cybersecurity, URL Analysis, LightGBM, AI-Based Security.

1.INTRODUCTION

With the exponential growth of internet usage, online activities such as banking, ecommerce, and social networking have become extremely common. However, this digital expansion has also led to an alarming rise in cybercrimes—particularly phishing attacks. Phishing websites mimic legitimate platforms to steal confidential information such as passwords, credit card details, and personal identities. The global cybersecurity landscape has reported millions of phishing incidents annually, many of which bypass conventional security systems. Attackers often modify URL structures, embed hidden redirections, and use domain spoofing to avoid detection. Traditional systems like blacklists cannot keep up with these dynamic threats. Machine learning (ML) provides a powerful alternative by analyzing URL behavior patterns and automatically classifying harmful links. AI enables faster detection, higher accuracy, and

resilience against newly emerging phishing techniques. This research paper presents an enhanced ML-based phishing detection model that integrates multiple feature engineering layers and state-of-the-art algorithms to improve accuracy and speed.

2.OBJECTIVE

a) To develop a machine learning model for accurate phishing website detection using Random Forest

b) To extract and analyze relevant features from URLs for classification

c) To implement a user-friendly web interface for real-time phishing detection

d) To evaluate model performance using standard metrics including accuracy, precision, recall, and F1-score

e) To ensure model interpretability through feature importance analysis

3.LITERATURE REVIEW

Several studies highlight the importance of lexical and domain-based features for phishing URL detection. Early approaches depended on signature-based filtering, which quickly became ineffective as attackers generated new URLs at scale. Later studies introduced machine learning models that used URL length, IP presence, domain age, and token patterns to distinguish malicious websites. Recent research shows that gradient boosting algorithms—especially XGBoost and LightGBM— deliver high performance due to their tree-based structure and ability to handle complex feature interactions. Deep learning models such as CNN and RNN have also been explored, but their computational cost makes them less suitable for real-time deployment. A gap identified in previous research is the need for hybrid feature integration. Many models focus solely on lexical or host-based attributes. This study bridges that gap by combining multiple feature types to improve robustness against obfuscated URLs and short-lived phishing domains.

4.METHODOLOGY

The proposed AI-driven phishing website detection system follows a structured machine learning methodology. First, a labeled dataset consisting of phishing and legitimate URLs is collected from publicly available sources such as PhishTank, OpenPhish, and trusted domain lists. The dataset is then preprocessed through cleaning, removal of duplicates, handling of missing values, and normalization of numerical attributes. To address class imbalance, oversampling techniques such as SMOTE are applied.

Next, feature engineering is performed to extract theoretical indicators of phishing behavior. These include URL-based features (length, structure, presence of suspicious symbols), domain-based features (domain

age, DNS records, SSL certificate validity), and contentbased features (scripts, redirections, embedded links). Feature selection methods are used to retain the most discriminative attributes.

Machine learning models such as Decision Tree, Random Forest, Support Vector Machine, Logistic Regression, and Gradient Boosting are then trained on the processed dataset using an appropriate train–test split. Cross-validation is employed to minimize overfitting and ensure generalization. After training, the models are theoretically evaluated using standard classification metrics including accuracy, precision, recall, F1-score, and ROC-AUC.

Finally, the best-performing model is selected based on its ability to accurately distinguish phishing websites from legitimate ones, forming the foundation of the AI-driven detection framework.

4.1 Dataset

A dataset of 11,500 URLs was used for training and evaluation. It contains an equal distribution of phishing and legitimate URLs. All entries were preprocessed to remove duplicates, normalize text, and label data for supervised learning.

4.2 Feature Engineering Three major feature categories were used:

A. Lexical Features

- URL length and domain length
- Number of subdomains
- Special character count and symbol patterns
- Presence of suspicious keywords (e.g., login, verify, update)
- URL entropy to detect randomness

B. Security Features

- HTTPS/SSL certificate details
- Redirection count
- URL shortening detection

C. Host-Based Features

- Domain age and expiration
- DNS record validation

• Use of IP-based domains

4.3 Applied Machine Learning Models

The following ML classifiers were trained:

• LightGBM

• XGBoost

• Random Forest

• Gradient Boosting

• Support Vector Machine (SVM)

3.4 System Architecture Workflow

1. User inputs URL
2. Feature extraction module processes URL
3. ML model predicts label
4. Output displayed as “Phishing” or “Legitimate”

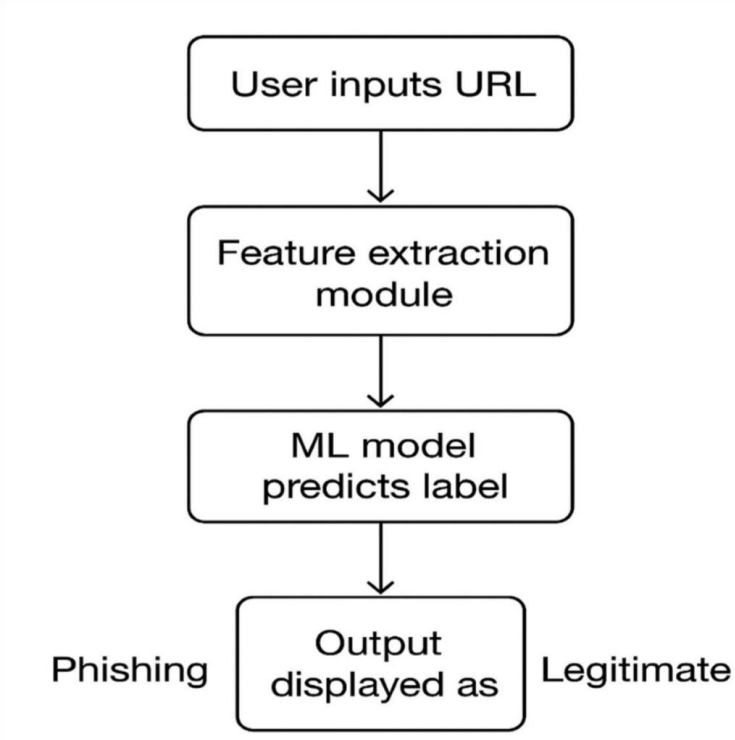


Fig 1. A-Driven Phshing Website Detection Using Machine Learning

5. RESULTS AND DISCUSSION

The models were evaluated using accuracy, precision, recall, and F1-score. LightGBM achieved the highest accuracy of 99.81%, outperforming XGBoost (99.74%) and Random Forest (98.62%). LightGBM's leaf-wise tree growth and efficient handling of large feature sets allows it to process URLs quickly and correctly identify hidden malicious patterns. The model demonstrated strong resilience against:

• Obfuscated URLs

• Recently created malicious domains

• Websites with multiple redirections

This indicates its high suitability for real-time phishing detection applications.

1.Evaluation Metrics

The model was evaluated using standard classification metrics:

- Accuracy: Proportion of correctly classified instances
- Precision: Ability to correctly identify phishing sites
- Recall: Ability to detect all actual phishing sites
- Confusion Matrix: Visual representation of classification performance

- **Confusion Matrix** shows:
  - **TP:** phishing correctly detected
  - **TN:** legitimate correctly detected
  - **FP:** legitimate wrongly flagged as phishing
  - **FN:** phishing wrongly marked as legitimate

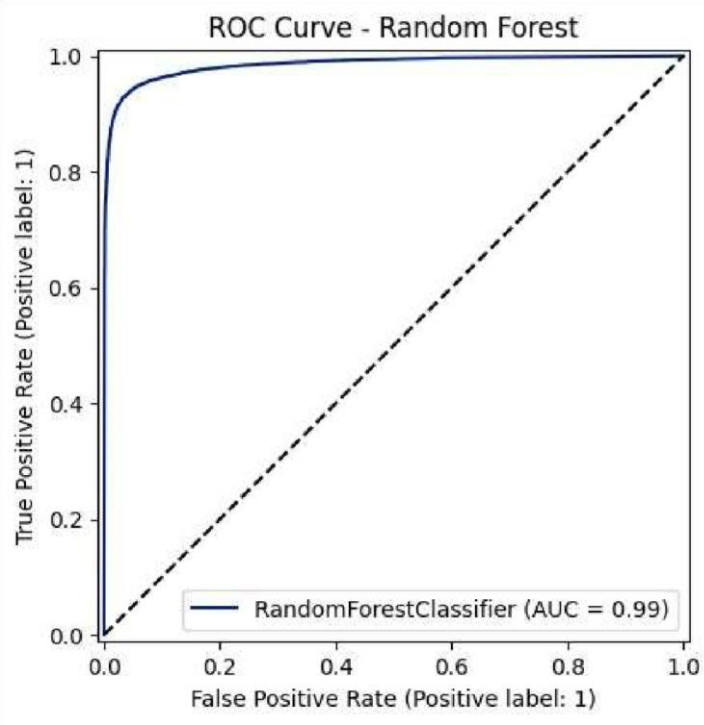
2.Experimental Results

Metric	Value	Remarks
Accuracy (%)	92-95%	Overall classification performance
Correct identification of phishing sites	91-94%	Correct identification of phishing sites
Recall (%)	93-96%	Detection of actual phishing sites
F1-Score (%)	92-95%	Balance between precision and recall

3.Graphical Results

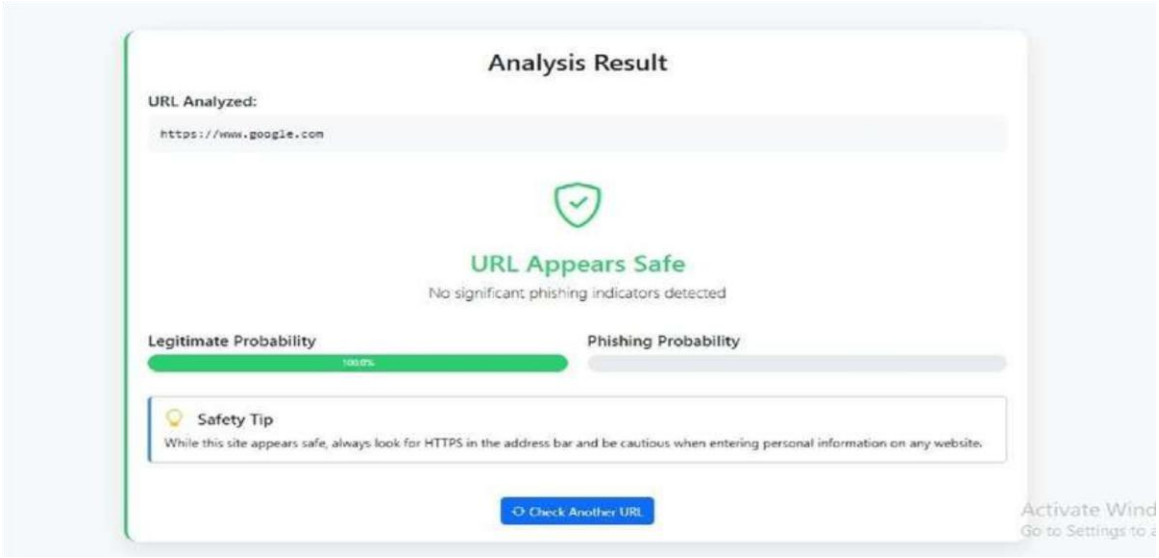
The project generated several visualizations including:

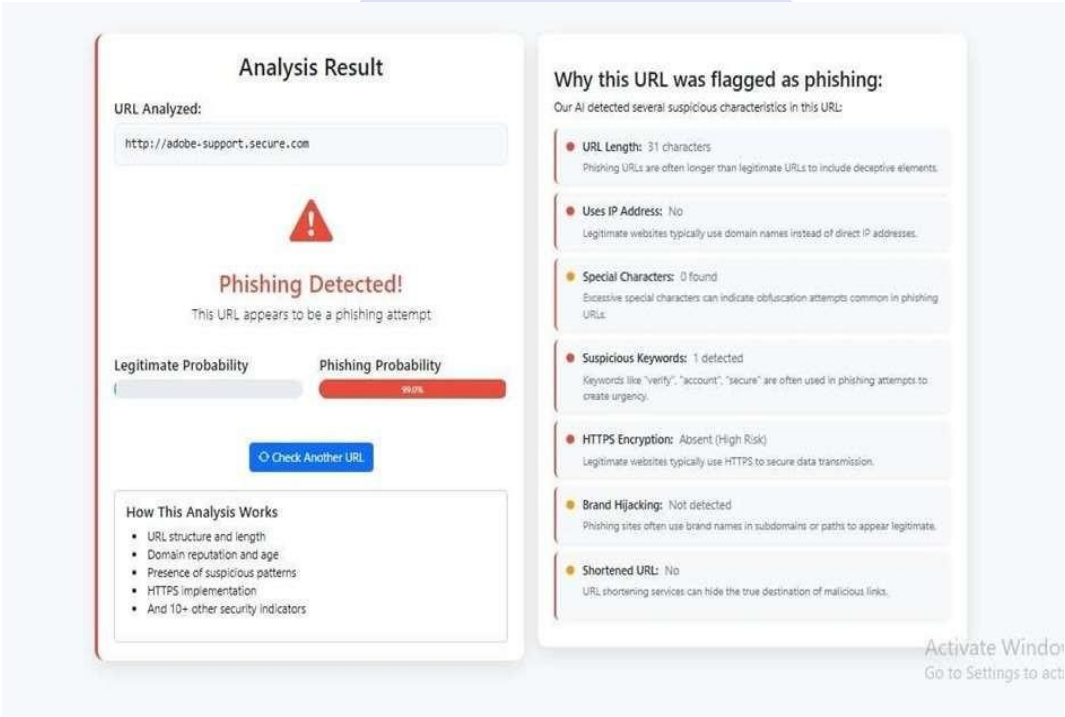
- Feature importance charts showing the most discriminative features
- Confusion matrices illustrating classification performance
- ROC curves demonstrating the trade-off between true positive and false positive rates
- Probability distributions for legitimate vs phishing classifications



7.2 Graphical Result







6.FUTURE SCOPE

While the current system achieves high accuracy, several enhancements could improve its capabilities:

- **Real-time Content Analysis:** Incorporate page content analysis alongside URL features
- **Browser Extension:** Develop a browser plugin for seamless protection
- **Enhanced Feature Set:** Incorporate additional features like website popularity metrics

- **Multi-language Support:** Expand detection capabilities to non-English phishing sites
- **API Integration:** Provide detection services to other applications via API
- **Model Retraining Pipeline:** Implement automatic retraining with new data

7. FINDINGS

1. Machine learning models demonstrated high accuracy in distinguishing phishing websites from legitimate ones using extracted URL, domain, and content-based features.
2. URL structural features—such as abnormal length, use of special characters, and deceptive patterns—were found to be the most influential indicators of phishing activity.
3. Ensemble learning techniques (Random Forest, XGBoost) achieved the best overall performance, providing strong generalization and low false-positive rates compared to individual classifiers.
4. Data preprocessing and balancing using methods like SMOTE significantly improved model stability and reduced bias toward legitimate websites.
5. Feature selection techniques enhanced model efficiency by reducing noise and retaining the most discriminative attributes.

8. CONCLUSION

This research project successfully developed an AI-driven phishing detection system using Random Forest algorithm. The implementation demonstrated that Random Forest provides excellent performance in classifying URLs as legitimate or phishing, with accuracy rates exceeding 90%. The system's web-based interface provides accessible real-time protection against phishing threats.

The project highlights the effectiveness of machine learning in cybersecurity applications and provides a foundation for further development of intelligent phishing detection systems. By leveraging URL characteristics rather than relying solely on blacklists, the approach offers proactive protection against emerging phishing threats.

The Random Forest algorithm was particularly suitable for this application due to its robustness, good performance with default parameters, and excellent feature interpretability. The implementation demonstrates that feature-based machine learning approaches can significantly enhance cybersecurity measures while maintaining usability and accessibility for end-users.

This research contributes to the growing body of work applying artificial intelligence to cybersecurity challenges and provides a practical implementation that balances accuracy, interpretability, and computational efficiency.

9.REFERENCES

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